# Malaria detection from cell images

Spritz++

May 25, 2025

## 1 Introduction

Malaria is a life-threatening infectious disease caused by the Plasmodium parasite, transmitted through the bite of infected Anopheles mosquitoes. It is most prevalent in subtropical and tropical regions such as Latin America, Asia, and Africa, which carry the heaviest burden, contributing to 94% of the estimated global cases and 95% of the estimated deaths. Malaria diagnosis relies on the microscopic examination of blood smears by trained technicians, which requires minimal and common lab infrastructure such as microscopes and solvents.

## 2 Goal

The primary goal of this project is to develop an automated detector capable of identifying malaria-infected blood cells from digitized microscopic images.

The provided input material is made of thin blood-smear microscopy images, each one comes with a ground truth .txt file containing all of the single blood-cells information, such as an ID number, cell type (which will be the searched label), spatial position, and furthermore. More information about these data is provided in the "ReadMe.pdf", which is part of the original dataset; only the "Polygon Set" type logs are meant to be used in this work, and ID numbers of the patients are also not required in this project.

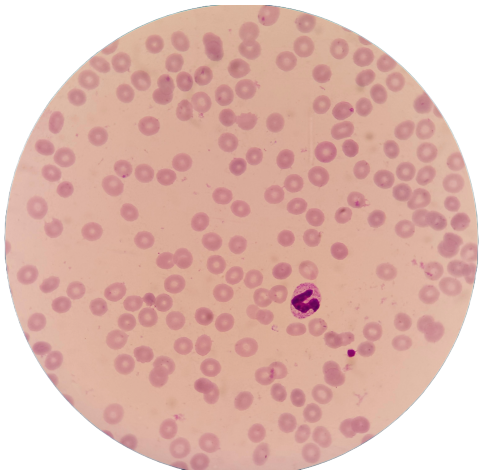


Figure 1: Type of input image (given in the dataset)

Since the scope is not limited to single-cell binary labeling, the program will have to process a whole image, identifying all the single cells, giving their locations, and labeling them.

The detector should produce as output a text report with all the localized cells and relative predicted labels, so indicating whether the cell is "Infected" or "Uninfected". The program should also produce as output one or more images with a meaningful results representation, showing the result of only a single or many tasks, e.g., one image with all the bounding box highlighting of the successfully localized cells, one image that does the same for all the parasitized cells, etc.

More details: parasites could be confused with some healthy cell purple imperfection; these cases should be handled carefully. Also, sometimes a few white blood cells are present in the images; the detector should not consider them.

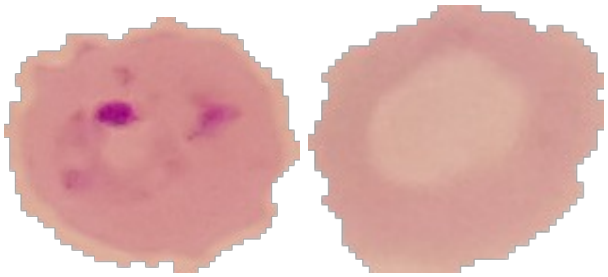


Figure 2: Parasitized (left) and Uninfected (right) single-blood cell example (both not given in the dataset)

# 3 Limitations (what is not required/requested)

Any classification of parasite species is not required: this project focuses only on binary classification, thus infected vs. uninfected status. Giving additional comments (other than labeling) about each single cell (as reported in the ground truths' data) is not requested for this task.

# 4 Possible Applications

Developing an automated malaria detector from cell images addresses several real-world needs and can be leveraged in multiple contexts: reference laboratories and research centers often audit local clinic diagnoses to ensure accuracy. Human error or fatigue can lead to false negatives in low parasitemia samples; in those scenarios, a software-based check can highlight possible wrong classifications, improving overall diagnostic reliability.

Researchers frequently analyze large repositories of digitized blood smear images to study parasite morphology or drug resistance trends. An automated classifier can rapidly label historical datasets, speeding up researchers' work.

# 5 Performance measurement

We will use the mean accuracy metric to quantify how well the malaria detector discriminates between infected and uninfected cells. The mean IoU is used to measure each blood cell detection.

## 5.1 Dataset structuree

Our Malaria_Dataset structure:

- GT: A collection of all the ground truth image-related text files
- Img: A set of blood smear microscopy images
- ReadMe.pdf: original dataset info file

All the proposed images and relative ground truths have been chosen from the dataset (National Library of Medicine, National Institutes of Health, Bethesda, MD, USA) provided by the paper "KASSIM, Yasmin M., et al. Clustering-based dual deep learning architecture for detecting red blood cells in malaria diagnostic smears. ieee journal of biomedical and health informatics, 2020, 25.5: 1735-1746.", available at https://data.lhncbc.nlm.nih.gov/public/Malaria/NIH-NLM-ThinBloodSmearsPf/index.html.